NexusRAG: Verifiable Retrieval-Augmented Architecture

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1. Abstract and Overview

Modern document research demands complete transparency, auditability, and exact sentence-level traceability. Traditional retrieval architectures suffer from context dilution, ungrounded extrapolation, and disconnected vector references. NexusRAG addresses these challenges through a dual-stage LangGraph pipeline that integrates deterministic query parsing, high-recall vector candidate selection, cross-scoring relevance reranking, and server-side citation validation.

2. Document Ingestion and Bounding-Box Geometry

Incoming documents in PDF, Markdown, and TXT formats undergo character-offset parsing using PyMuPDF. For every extracted sentence, NexusRAG computes exact pixel bounding boxes (x0, y0, x1, y1) alongside sequential character spans. This structural metadata is indexed into FAISS alongside dense semantic vectors generated by Google Gemini embedding models (768-dimensional embeddings).

When queries are processed, the system retrieves top-k candidates, reranks them using neural relevance cross-encoders, and computes an aggregate evidence sufficiency score before synthesis begins.

3. LangGraph Execution Workflow

The synthesis engine executes a nine-node Directed Acyclic Graph. The workflow begins with input validation and query rewriting to produce a standalone search term. Next, FAISS performs dense retrieval over the isolated tenant index, yielding candidate passages. The reranking node scores candidate alignment against the rewritten query. If the evidence sufficiency metric exceeds the calibrated threshold of 0.650, synthesis proceeds to the grounded generation node.

4. Server-Side Citation Validation and Anti-Hallucination Controls

A dedicated post-processing gate audits every citation emitted by the generation model. Bracketed references [1], [2] are parsed against the active retrieved candidate set. Any hallucinated citation index pointing to unretrieved text is stripped server-side prior to client dispatch. Every surviving citation links deterministically to the database chunk and page offset.